

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the paper.
2. You must NOT start writing during reading time.
3. The question paper has 9 printed pages.
4. It consists of twenty questions and four sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in three questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner

Kindly read aloud the instructions given above to all the candidates present in the examination hall.

SECTION A – 20 MARKS**Question 1**

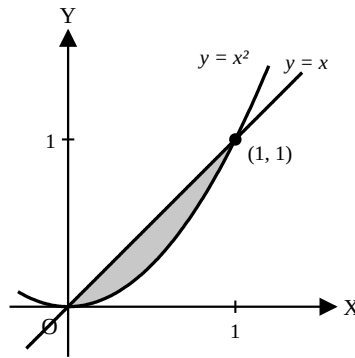
In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) Let $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)\}$ be a relation on the set $\{1, 2, 3\}$. Then R is: [1]
(Understand)
- (a) reflexive and transitive
 - (b) reflexive but neither symmetric nor transitive
 - (c) an equivalence relation
 - (d) symmetric and transitive

- (ii) If α and β are respectively the least and the greatest values of $2 \sin^{-1} x + \cos^{-1} x$ for $x \in [-1, 1]$, then: (*Analyse*) [1]
- (a) $\alpha = -\pi/2, \beta = \pi/2$
- (b) $\alpha = 0, \beta = \pi$
- (c) $\alpha = -\pi, \beta = \pi$
- (d) $\alpha = \pi/4, \beta = 3\pi/4$
- (iii) The value of the determinant $\begin{vmatrix} a-b & b-c & c-a \\ b-c & c-a & a-b \\ c-a & a-b & b-c \end{vmatrix}$ is: (*Understand*) [1]
- (a) 0
- (b) abc
- (c) $a + b + c$
- (d) $(a-b)(b-c)(c-a)$
- (iv) The volume of a cube is increasing at the rate of $9 \text{ cm}^3/\text{s}$. How fast is its surface area increasing when the length of an edge is 10 cm ? (*Apply*) [1]
- (a) $3.6 \text{ cm}^2/\text{s}$
- (b) $1.8 \text{ cm}^2/\text{s}$
- (c) $36 \text{ cm}^2/\text{s}$
- (d) $0.36 \text{ cm}^2/\text{s}$
- (v) Statement I: $f(x) = \tan x$ is strictly increasing on $(-\pi/2, \pi/2)$. [1]
Statement II: $g(x) = \sin x$ is strictly increasing on $(0, \pi)$.
Which of the following is correct? (*Understand*)
- (a) Statement I is true and Statement II is false.
- (b) Statement I is false and Statement II is true.
- (c) Both the statements are true.
- (d) Both the statements are false.
- (vi) The slope of the normal to the curve $y = 2x^2 + 3 \sin x$ at $x = 0$ is: (*Apply*) [1]
- (a) 3
- (b) $1/3$
- (c) -3
- (d) $-1/3$
- (vii) For $f(x) = x^3 - 3x^2 + 3x + 1$, $f'(1) = 0$ and $f''(1) = 0$. Which of the following is correct? [1]
(*Analyse*)
- (a) f has a local maximum at $x = 1$.
- (b) f has a local minimum at $x = 1$.
- (c) f has neither a local maximum nor a local minimum at $x = 1$, since f' does not change sign there.
- (d) f has a local minimum at $x = 1$, because $f''(1) = 0$.

(viii) The expression for the area of the shaded region bounded by $y = x$ and $y = x^2$ is: (*Apply*)

[1]



(a) $\int_0^1 (x^2 - x) dx$

(b) $\int_0^1 (x - x^2) dx$

(c) $\int_0^1 (y - \sqrt{y}) dy$

(d) $\int_0^2 (x - x^2) dx$

(ix) Shown below is a solution of $I = \int_0^{\pi/2} \frac{\cos x}{1 + \sin x} dx$:

[1]

Step 1 Put $t = 1 + \sin x$, so $dt = \cos x dx$.

Step 2 When $x = 0$, $t = 1$; when $x = \pi/2$, $t = 2$. So $I = \int_1^2 \frac{1}{t} dt$.

Step 3 $I = [\log t]_1^2 = \log 2 - 1$.

In which step is there an error (if any)? (*Understand*)

(a) Step 1

(b) Step 2

(c) Step 3

(d) No error

(x) On evaluating $\int e^{\log x} dx$, Aman's answer was $e^{\log x} + c$ and Bela's answer was $x^2/2 + c$. Whose answer is correct? (*Understand*)

[1]

(a) Only Aman's answer is correct.

(b) Only Bela's answer is correct.

(c) Both of them are correct.

(d) Neither of them is correct.

(xi) The value of λ for which the vectors $2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular is: (*Apply*)

[1]

(a) $-5/2$

(b) $5/2$

(c) $2/5$

(d) 5

- (xii) The direction cosines of the line joining $P(1, 2, 3)$ and $Q(-1, 0, 4)$, directed from P to Q , are: [1]
(Understand)
 (a) $-2/3, -2/3, 1/3$
 (b) $-2, -2, 1$
 (c) $2/3, 2/3, 1/3$
 (d) $-1/3, -1/3, 2/3$
- (xiii) The objective function $Z = px + qy$ ($p, q > 0$) of an LPP has the same maximum value at the corner points $(3, 4)$ and $(0, 5)$. Then: *(Apply)* [1]
 (a) $p = q$
 (b) $q = 3p$
 (c) $p = 3q$
 (d) $3p = 4q$
- (xiv) Statement I: A corner point of a feasible region is a point of the region which is the intersection of two of its boundary lines. [1]
 Statement II: If the feasible region of an LPP is unbounded, a maximum value of the objective function may not exist.
 Which of the statements is correct? *(Understand)*
 (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (xv) A and B are independent events with $P(A) = 0.3$ and $P(B) = 0.4$. Then $P(A \cap B')$ is: *(Apply)* [1]
 (a) 0.12
 (b) 0.18
 (c) 0.28
 (d) 0.42
- (xvi) Assertion: The number of bijective functions from $A = \{1, 2, 3\}$ to itself is 6. [1]
 Reason: The number of one-one functions from a set with n elements to itself is $n!$.
 Which one of the following is correct? *(Understand)*
 (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.

(xvii) Assertion: $\int_{-\pi}^{\pi} \sin^3 x \cos^2 x \, dx = 0$. [1]

Reason: If f is an odd continuous function on $[-a, a]$, then $\int_{-a}^a f(x) \, dx = 0$.

Which one of the following is correct? (*Understand*)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xviii) If the matrix $\begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & x \\ 3 & -4 & 0 \end{bmatrix}$ is skew-symmetric, find the value of x . (*Understand*) [1]

(xix) Find the integrating factor of the differential equation $x \frac{dy}{dx} - y = x^2 \sin x$, $x > 0$. (*Apply*) [1]

(xx) A random variable X has the distribution $P(X = 0) = k$, $P(X = 1) = 2k$, $P(X = 2) = 3k$, $P(X = 3) = 4k$. Find $P(X \geq 2)$. (*Apply*) [1]

SECTION B – 14 MARKS

Question 2

Show that the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{5} = \frac{y-1}{2} = z$ intersect. Find their point of intersection. (*Apply*) [2]

Question 3

Three archers A, B and C hit a target with probabilities $1/2$, $2/3$ and $3/4$ respectively, independently of one another. If each shoots once, find the probability that exactly two of them hit the target. (*Apply*) [2]

Question 4

Solve the differential equation $\frac{dy}{dx} + y \cot x = 2 \cos x$, given that $y = 0$ when $x = \pi/2$. (*Apply*) [2]

Question 5

Find the values of a and b such that the function [2]

$$f(x) = \begin{cases} 5, & x \leq 2 \\ ax + b, & 2 < x < 10 \\ 21, & x \geq 10 \end{cases}$$

is continuous. (*Understand*)

OR

Show that $f(x) = |x - 3|$ is continuous but not differentiable at $x = 3$. (*Understand*) [2]

Question 6

Solve for x : $\sin^{-1}(1-x) - 2 \sin^{-1} x = \pi/2$. (Apply) [2]

OR

Prove that $\tan^{-1}(1/2) + \tan^{-1}(1/5) + \tan^{-1}(1/8) = \pi/4$. (Apply) [2]

Question 7

Find the area of the triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$ and $C(1, 5, 5)$. (Apply) [2]

OR

If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $|\vec{a} + \vec{b}| = \sqrt{7}$, find the angle between $\vec{a}$ and $\vec{b}$, and $|\vec{a} - \vec{b}|$. (Apply) [2]

Question 8

During a light show three drones hover at $A(2, 6, 3)$, $B(1, 2, 7)$ and $C(3, 10, -1)$, with distances in metres.

- (i) Find $\vec{AB}$ and $\vec{BC}$. (Understand) [1]
- (ii) Show that the three drones are in a straight line and state which drone is between the other two. (Analyse) [1]

SECTION C – 21 MARKS**Question 9**

A dealer sells table fans and ceiling fans. A table fan costs him ₹1,000 and a ceiling fan ₹500. He has ₹50,000 to invest and space to store at most 60 fans. He makes a profit of ₹100 on a table fan and ₹60 on a ceiling fan. Assuming he sells all the fans he buys:

- (i) Formulate the problem as an LPP to maximise his profit. (Apply) [1]
- (ii) Draw the feasible region and find its corner points. (Apply) [1]
- (iii) Find the number of fans of each type he should buy for maximum profit. (Evaluate) [1]

Question 10

If $y = Ae^{mx} + Be^{nx}$, where A, B are constants, prove that $\frac{d^2y}{dx^2} - (m+n)\frac{dy}{dx} + mny = 0$. [3]
(Apply)

Question 11

A curve is given by $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$, $0 < t < \pi/2$.

- (i) Find $\frac{dy}{dx}$. (Apply) [2]
- (ii) Hence find $\frac{d^2y}{dx^2}$. (Analyse) [1]

Question 12

Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and $\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$. (Analyse) [3]

OR

P is the point (1, 3, 4) and π is the plane $2x - y + z + 3 = 0$.

- (a) Write the equation of the line through P perpendicular to π . (Understand) [1]
 (b) Find the foot of the perpendicular from P to π . (Apply) [1]
 (c) Find the image of P in π . (Apply) [1]

Question 13

If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 4A + 3I = O$. Hence find A^{-1} . (Apply) [3]

OR

Solve for x, given $a \neq 0$:

$$\begin{vmatrix} x+a & x & x \\ x & x+a & x \\ x & x & x+a \end{vmatrix} = 0.$$

(Analyse)

Question 14

Let $I = \int_1^4 \frac{\sqrt{x}}{\sqrt{x} + \sqrt{5-x}} dx$.

- (i) Using $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, show that $I = \int_1^4 \frac{\sqrt{5-x}}{\sqrt{5-x} + \sqrt{x}} dx$. (Understand) [1]
 (ii) Hence evaluate I. (Apply) [2]

Question 15

Using integration, find the area of the region enclosed between the parabolas $y^2 = 2x$ and $x^2 = 2y$. (Apply) [3]

OR

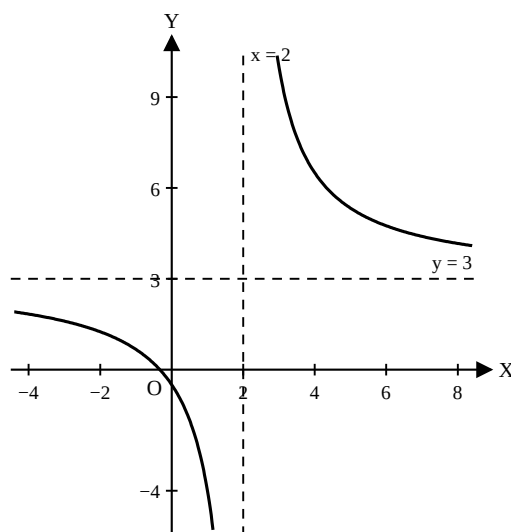
Consider the parabola $y = x^2 + 1$ and the line $x + y = 3$.

- (a) Find their points of intersection. (Understand) [1]
 (b) Using integration, find the area of the region enclosed between them. (Apply) [2]

SECTION D – 25 MARKS

Question 16

A function $f : \mathbb{R} - \{2\} \rightarrow \mathbb{R} - \{3\}$ is defined by $f(x) = \frac{3x+1}{x-2}$. Its graph is shown below.



- | | | |
|-------|--|-----|
| (i) | Show that f is one-one. (<i>Understand</i>) | [1] |
| (ii) | Show that f is onto. (<i>Understand</i>) | [1] |
| (iii) | Find $f^{-1}(x)$. (<i>Apply</i>) | [1] |
| (iv) | Verify that $(f \circ f^{-1})(4) = 4$. (<i>Apply</i>) | [1] |
| (v) | Find the values of x for which $f(x) = f^{-1}(x)$. (<i>Analyse</i>) | [1] |

Question 17

Water is poured at the rate of $3 \text{ m}^3/\text{min}$ into a tank in the shape of an inverted right circular cone of height 10 m and top radius 5 m.

- | | | |
|-----|--|-----|
| (a) | Find the rate at which the water level is rising when the depth is 4 m. (<i>Apply</i>) | [2] |
| (b) | Show that the level rises more and more slowly as the tank fills. (<i>Evaluate</i>) | [1] |
| (c) | Find the rate at which the area of the water surface is increasing when the depth is 4 m. (<i>Apply</i>) | [2] |

OR

An open tank with a square base and vertical sides is to hold 32 m^3 of water. Sheet metal for it costs ₹500 per m^2 .

- | | | |
|-----|--|-----|
| (a) | If the side of the base is x m, express the area S of sheet used as a function of x . (<i>Apply</i>) | [2] |
| (b) | Find $S'(x)$. (<i>Understand</i>) | [1] |
| (c) | Find the dimensions of the tank for which the cost is least, and that cost. (<i>Evaluate</i>) | [2] |

Question 18

- | | | |
|-----|---|-----|
| (a) | Resolve $\frac{1}{(x+1)(x+2)}$ into partial fractions and hence find $\int \frac{1}{(x+1)(x+2)} dx$. (<i>Understand</i>) | [2] |
|-----|---|-----|

- (b) Hence evaluate $\int \frac{\sec^2 x}{(\tan x + 1)(\tan x + 2)} dx$. (Evaluate) [3]

OR

- (a) Prove that $\int_0^\pi x f(\sin x) dx = \frac{\pi}{2} \int_0^\pi f(\sin x) dx$. (Understand) [2]
- (b) Hence evaluate $\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$. (Evaluate) [3]

Question 19

Bag I contains 3 red and 4 black balls and Bag II contains 5 red and 6 black balls. A bag is chosen at random and a ball is drawn from it.

- (a) Find the probability that the ball drawn is red. (Apply) [2]
- (b) The ball drawn is red. Find the probability that it came from Bag II. (Analyse) [2]
- (c) Which bag is the red ball more likely to have come from? (Evaluate) [1]

OR

A lot of 15 bulbs contains 5 defective bulbs. Two bulbs are drawn at random, one after the other, without replacement. Let X be the number of defective bulbs drawn.

- (a) Write the possible values of X. (Understand) [1]
- (b) Find the probability distribution of X. (Apply) [2]
- (c) Find the mean of X. (Apply) [2]

Question 20

The number of visitors (in hundreds) to a science fair on day x is modelled by $V(x) = ax^2 + bx + c$. There were 600 visitors on day 1, 1,100 on day 2 and 1,800 on day 3. Form the equations, write them as $AX = B$, solve using A^{-1} , and hence predict the number of visitors on day 5. (Evaluate) [5]

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